4 Time Response Analysis

INTRODUCTION

The time response of a control system means as to how a system behaves in accordance with time when a specified input test signal is applied.

- Time response analysis is a study of output with respect to time.
- By time response analysis we can study –
  1. Stability of the system
  2. Accuracy of the system
  3. Error in the system
- Time response of a control system is divided in two parts –
  1. Transient Response
  2. Steady State Response

Definition

Time domain analysis refers to the analysis of system performance in time which mean evolution of system output with time.

1. Transient Response

- It is part of time response that goes to zero as time tends to be large.
- It is a temporary response of system.
- Transient response tells about the nature of response (i.e. oscillatory) or nature of damping and also gives idea of speed of system.

2. Steady State Response

- Steady state response reveals the accuracy of control system.
- It deals with estimation of magnitude of steady state error between input and output.
- Steady state means a state of the output of a control system as time approaches infinity after initiation of the input.
- Time response of a control system for a specified input is given below –

Gray Matter Alert!!!

Variation of output is fast initially in transient response, and variation of output is slow in steady state response.

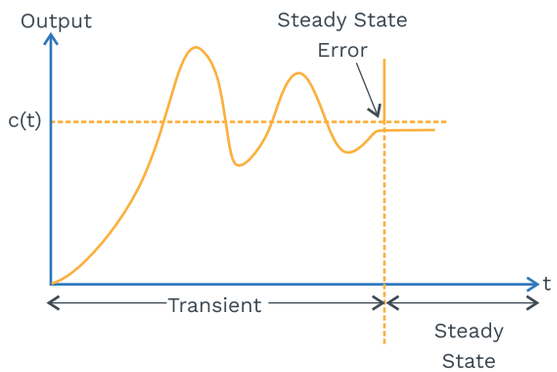


Figure-4.1. Time Response of a control system

Standard Test Signals

- There are basically four test signals which are applied for time response analysis of a control system. These are given below –

1. Step Signal → Sudden Input
2. Ramp Signal → Velocity type Input
3. Parabolic Signal → Acceleration type input

} Time Domain Analysis

4. Impulse Signal → Sudden Shocks } Stability Analysis

Rack your Brain

What are the parameters of transient response?

Time response of zero order control system

The equation describing a zero order system is

$$\frac{C(s)}{R(s)} = \frac{b_0}{a_0}$$

Let K = gain = $\frac{b_0}{a_0}$

$$\frac{C(s)}{R(s)} = K$$

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To evaluate transient state analysis and to define the transient state specification, the magnitude of the input signal is used.



Figure-4.2. Block diagram of zero order system

For example – Sensor, potentiometer, tachometers etc.
Graphically, we can plot the response as -

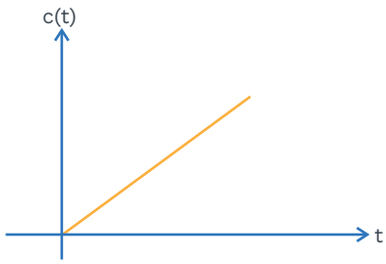


Figure-4.3.

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Time response analysis is not applicable to zero order system because input and output characteristics are linearly dependent.

Time Response of a first order system

A first order control system is expressed by a transfer function as –

$$\frac{C(s)}{R(s)} = \frac{1}{sT+1}$$

The block diagram representation of above expression is –

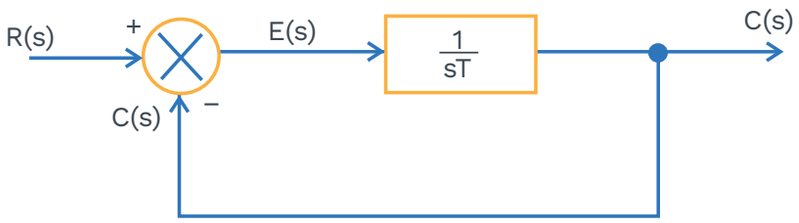


Figure-4.4. Block diagram of first order system

(a) Response to the unit step input –

If the input is unit step input –

$$r(t) = u(t) \text{ and } R(s) = \frac{1}{s}$$

Output response for unit step is –

$$C(s) = R(s) \cdot \frac{1}{sT+1}$$

$$C(s) = \frac{1}{s} \cdot \frac{1}{sT+1}$$

Taking the inverse Laplace transform, we get –

$$c(t) = 1 - e^{-t/T}$$

Which is plotted as

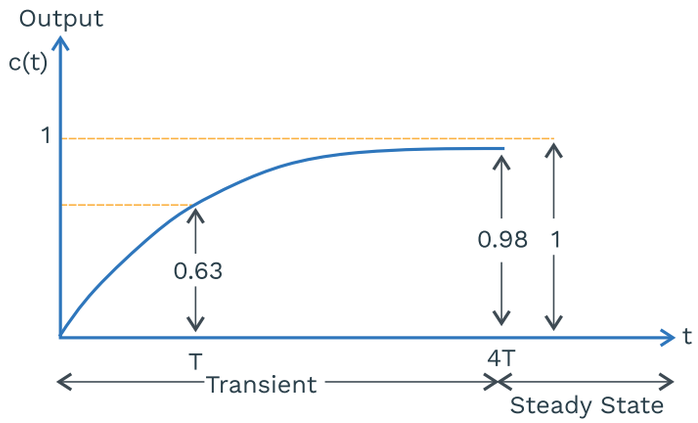


Figure-4.5. Unit step response of first order system

The output rises exponentially from zero value to final value of unity. Initial slope of curve at t = 0 is

$$\left.\frac{dc}{dt}\right|_{t=0} = \left.\frac{1}{T}e^{-t/T}\right|_{t=0} = \frac{1}{T}$$

Where, T = time constant of the system.
The error is given by,

$$e(t) = r(t) - c(t)$$

$$= 1 - \left(1 - e^{-t/T}\right)$$

$$= e^{-t/T}$$

The steady state error is,

$$e_{ss} = \lim_{t\to\infty} e(t) = 0$$

$$e_{ss} = 0$$

Thus, as the time increases the difference between input and output approaches to zero. Hence, the steady state error is zero.

(b) Response to unit ramp input –

$$C(s) = R(s) \cdot \frac{1}{sT+1}$$

As the input is unit ramp function

$$r(t) = t\,u(t) \text{ and } R(s) = \frac{1}{s^2}$$

Thus, the output response for unit ramp input is

$$C(s) = \frac{1}{s^2} \cdot \frac{1}{sT+1} = \frac{1}{s^2(sT+1)}$$

By using partial fraction

$$C(s) = \frac{1}{s^2} - \frac{T}{s} + \frac{T^2}{sT+1}$$

Taking the inverse Laplace transform, we get –

$$c(t) = t - T\left(1 - e^{-t/T}\right)$$

Definition

Time Constant is the time required for a system to reach 63.2% of [FV–IV].
Where,
FV=Final value,
IV= Initial value

Gray Matter Alert!!!

- The step response will not have any combination of system parameters.
- Examples of first order system are thermal system, pneumatic system etc.

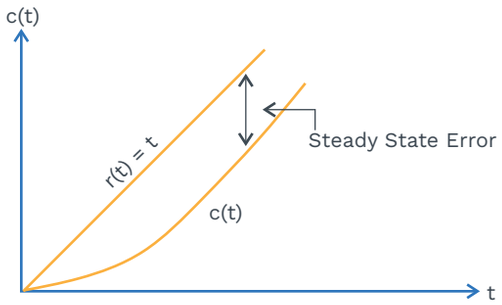


Figure-4.6. Unit ramp response of first order system

The error signal is,

$$e(t) = r(t) - c(t)$$

$$= t - \left(t - T + Te^{-t/T}\right)$$

$$e(t) = \left(T - Te^{-t/T}\right)$$

The steady state error is,

$$e_{ss} = \lim_{t\to\infty}\left(T - Te^{-t/T}\right)$$

$$e_{ss} = T$$

Thus, we can say that during steady state the output matches with the input but lags behind the input by time T.

Settling Time – The time taken by the output to settle near steady state value is called settling time.

Settling time for 2% tolerance band

The output must settle within 2% of steady state value.
Considering input as step input.

$$c(t) = A\left(1 - e^{-t/T}\right)$$

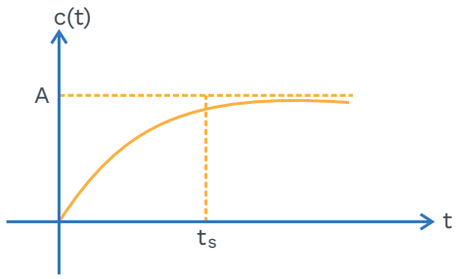


Figure-4.7

Steady state value = A
At $t = t_s, c(t) = 0.98A$

$$0.98A = A\left(1 - e^{-t_s/T}\right)$$

$$e^{-t_s/T} = 0.02$$

$$t_s = -T \ln 0.02 = 4T$$

$$t_s = 4T$$

Settling time = 4 times of the time constant, output settles within 2% of steady state.
Similarly, settling time for 5% tolerance band is-

$$t_s = 3T$$

Thus, we can conclude that –

Tolerance band	Settling Time (t_s)
2%	4T
5%	3T
1%	5T

Response to unit impulse input –
The output response for unit impulse input R(s) = 1 is given by –

$$C(s) = 1 \cdot \frac{1}{sT + 1}$$

Taking inverse Laplace transform

$$c(t) = \frac{1}{T} e^{-t/T}$$

Graphically, we can represent the above response as –

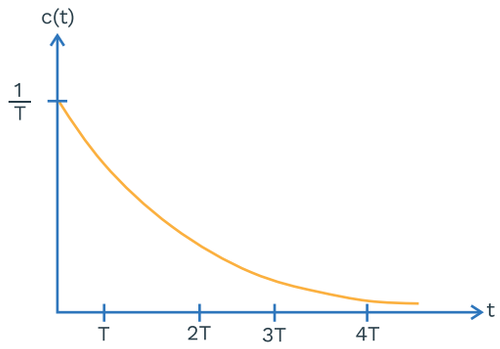


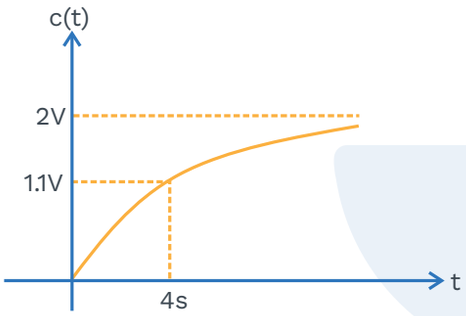
Figure-4.8. Unit impulse response of first order system

Example – 1

A first order system initially at rest is subjected to sudden input at t = 0, its response reaches 1.1V in 4 seconds and eventually reaches a steady state of 2V. Find the time constant.

Solution:

The response of first order system is,



$$c(t) = A\left(1 - e^{-t/T}\right)$$

$$\lim_{t\to\infty} c(t) = \lim_{t\to\infty} A\left(1 - e^{-t/T}\right) = A$$

And A = 2 (from the figure)

At $t = 4s, 1.1 = 2\left(1 - e^{-4/T}\right)$

$$2e^{-4/T} = 0.9$$

T = 5 seconds

Previous Year's Questions

For the system $\frac{2}{(s+1)}$, the approximate time taken for a step response to reach 98% of its final value is approximately

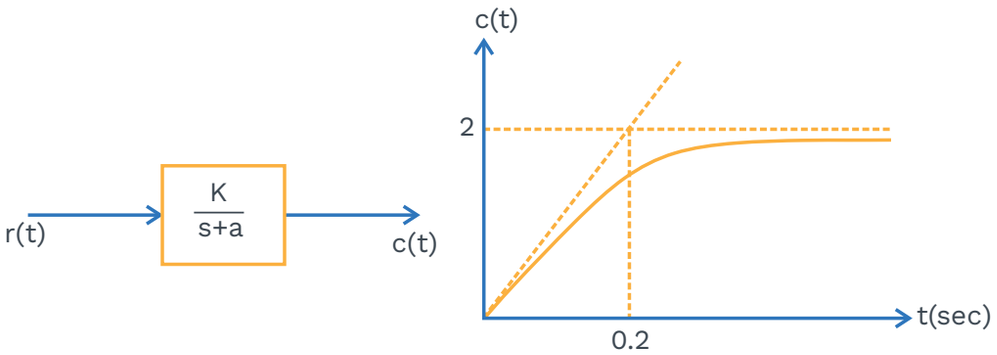
(a) 1 s (b) 2 s

(c) 4 s (d) 8 s

Ans. (c)

Example – 2

A first order system and its response to a unit step input are shown in figure below. Calculate system parameter K and a?



Solution:

$$\frac{C(s)}{R(s)} = \frac{K}{s+a} = \frac{K}{a\left(1+\frac{1}{a}s\right)}$$

$$T = \frac{1}{a}$$

From figure , time constant (T) = 0.2

$$\therefore \frac{1}{a} = 0.2$$

$$a = \frac{1}{0.2} = 5$$

Final value $= \lim_{s\to 0} s.C(s)$

$$= \lim_{s\to 0} s \cdot \frac{K}{s(s+a)}$$

$$= \frac{K}{a}$$

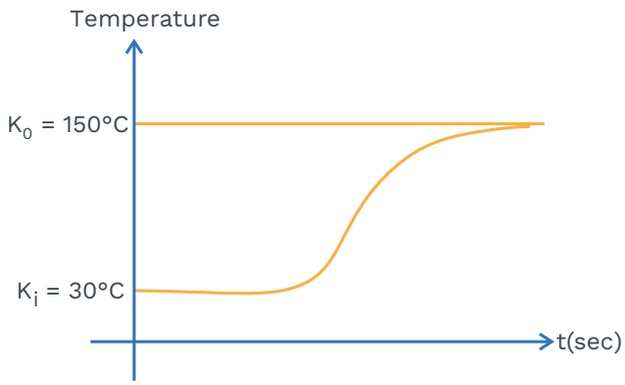
And $\frac{K}{a} = 2 \Rightarrow \frac{K}{5} = 2$

$K = 10$

Example – 3

A thermometer having first order dynamics is subjected to sudden temperature change of 30°C – 150°C. If it has a time constant at 4 sec, what temperature it will indicate after 4 seonds.

Solution:



$$c(t) = K_0 + (K_i - K_0)e^{-t/T}$$

$T = 4$ sec, $K_0 = 150^\circ C$, $K_i = 30^\circ C$

At t = 4 sec.

$$c(t) = 150 + (30 - 150)e^{-4/4}$$

$$c(t) = 150 - 44.14 = 105.85$$

Example – 4

Find the settling time for the 2% error band. If input is

$$r(t) = 10\,u(t) \text{ and } G(s) = \frac{1}{s+2}.$$

Solution:

$$G(s) = \frac{1}{s+2} = \frac{1}{2\left(1+\frac{1}{2}s\right)}$$

$$\therefore T = \frac{1}{2}$$

We know that

$$c(t) = A[1 - e^{-t\backslash T}]$$

For 2% error band, c(t) = 98% of A

$$\therefore 0.98A = A\left[1 - e^{-t_s/T}\right]$$

$$e^{-t_s/T} = 0.02$$

$$\frac{t_s}{T} = 3.91$$

$$t_s = 3.91\,T = \frac{3.91}{2} = 1.95$$

$$t_s = 1.95\,\text{sec}$$

Second Order System

- Second order system is the lowest order system capable of an oscillatory response to step-input For example – Spring mass damper system, RLC circuit etc.
- To analyse second order system two parameters are needed that is damping ratio (ξ) and undamped natural frequency (ω_n).

Definition

A second order system is one wherein the highest power of s in the denominator of its transfer function is 2.

Consider, second order system with unity feedback represented by following diagram –

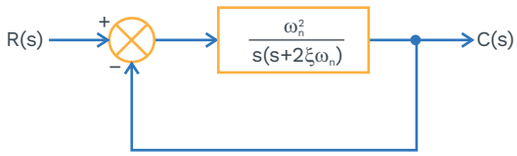


Figure-4.9. Block diagram of second order system

Open loop control system of the system is

$$G(s) = \frac{\omega_n^2}{s\left(s + 2\xi\omega_n\right)}$$

Standard form of closed loop transfer function is –

$$TF = \frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s) \cdot H(s)}$$

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\xi\omega_n s + \omega_n^2}$$

Where, ω_n = System Natural Frequency

ξ = Damping Ratio

For example – PMMC type indicating instruments

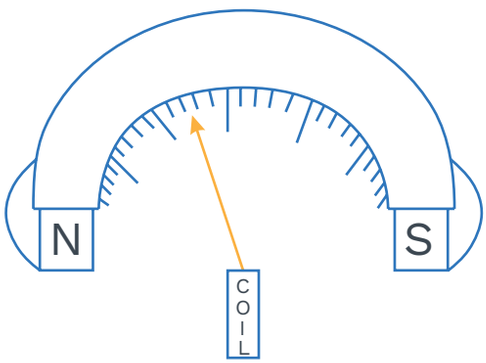


Figure-4.10.

Input = Deflecting torque (T_d)

Output = Angular Deflection pointer (θ)

J = Moment of inertia

B = Friction

K = Spring Constant

$$T_d = J \cdot \frac{d^2\theta}{dt^2} + B \cdot \frac{d\theta}{dt} + K\theta$$

or $$T_d(s) = \left[s^2 J + B \cdot s + K \right] \theta(s)$$

$$\frac{\theta(s)}{T_d(s)} = \frac{1}{Js^2 + Bs + K}$$

Transfer function is:

$$\frac{\theta(s)}{T_d(s)} = \frac{1/J}{s^2 + \frac{B}{J}s + \frac{K}{J}}$$

Comparing with standard form of second order system

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\xi\omega_n s + \omega_n^2}$$

$$\Rightarrow \omega_n = \sqrt{\frac{K}{J}} \quad \text{and} \quad \xi = \frac{B}{2\sqrt{KJ}}$$

Damping Ratio (ξ)

- It is a dimensionless quantity describing the decay of oscillations during a transient response.
- It is the ratio of actual damping to critical damping of a system.

Natural Frequency (ω_n)

- Angular frequency at which system tends to oscillate in the absence of damping force.

Damped Frequency (ω_d)

- Angular frequency at which system tends to oscillate in the presence of damping force and it is given by –

$$\omega_d = \omega_n \sqrt{1 - \xi^2}$$

Effect of damping on nature of response

- Response of second order system mainly depends on the damping ratio ξ. Transfer function of a second order system is

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\xi\omega_n s + \omega_n^2}$$

Characteristic equation obtained by the denominator of the standard transfer function is zero.

$$s^2 + 2\xi\omega_n s + \omega_n^2 = 0$$

The roots of above equation are closed loop poles of the control system. These are,

$$s_1, s_2 = \frac{-2\xi\omega_n \pm \sqrt{(2\xi\omega_n)^2 - 4\omega_n^2}}{2}$$

$$= -\xi\omega_n \pm \omega_n\sqrt{\xi^2 - 1}$$

$$= -\xi\omega_n \pm \omega_d$$

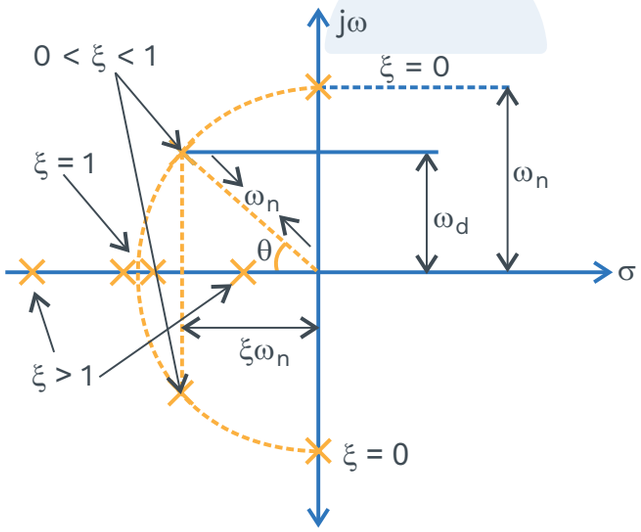
Real part of roots represents the damping factor and the imaginary part represents damped frequency of oscillations.

- For any test input, the response of 2nd order system can be studied in four cases depending on the damping effect created by value of ξ.
  1. $\xi > 1$; Over damped system
  2. $\xi = 1$; Critically damped system
  3. $0 < \xi < 1$; Underdamped system
  4. $\xi = 0$; Undamped system

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We do not consider negative damping ratio (ξ) because it means the oscillations are increasing in amplitude which results in unstable system.

Pole location of second order system on s-plane is as follows:



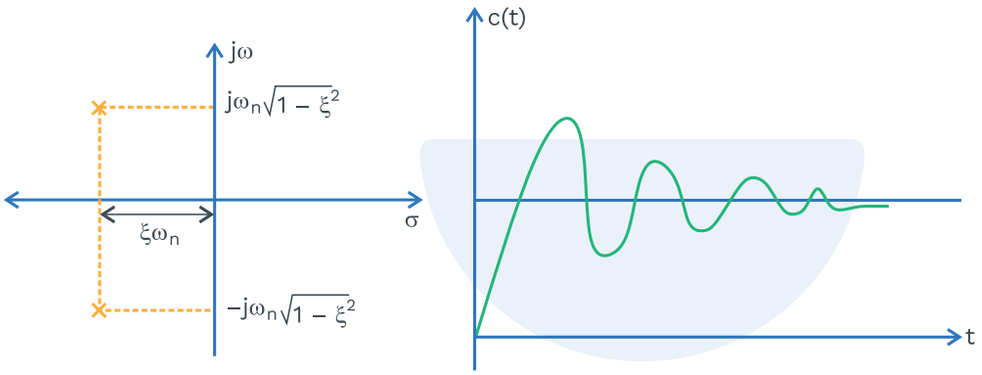
Figure–4.11. Pole location for second order system

System can be classified as follows depending on the range of damping ratio–

Case – 1: $0 < \xi < 1$

- Output response is under damped.
- System transients oscillate with the amplitude of oscillation gradually decreasing to zero.
- For under damped system, closed loop poles will be complex conjugate with negative real part.

Pole zero plot and output response for under damped system is given below.

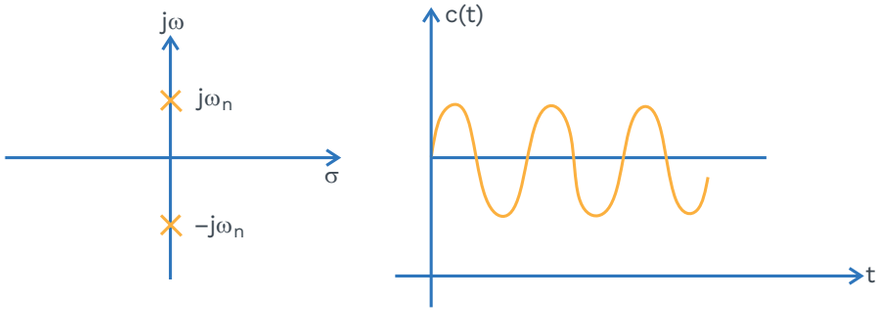


Figure–4.12.

Case – 2: $\xi = 0$

- Output response is undamped.
- System keeps on oscillating at its natural frequency without any decay in amplitude.
- Closed loop poles will be imaginary conjugate.

Location of roots and corresponding time response is given below.

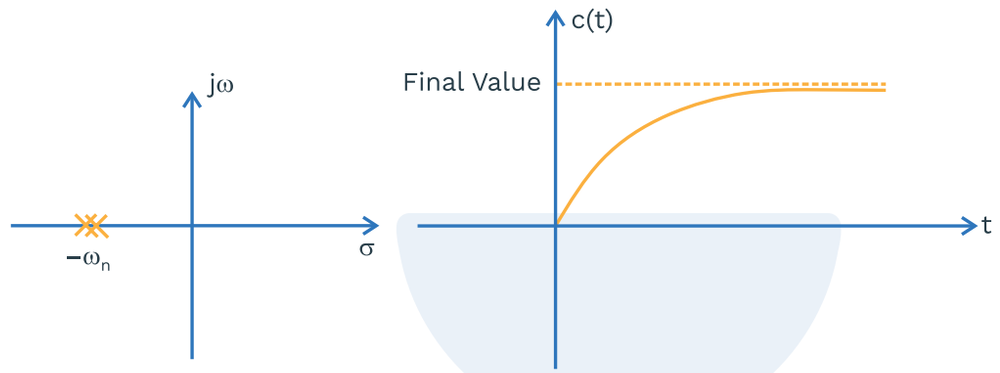


Figure–4.13.

Case – 3: $\xi = 1$

- Transient in the system decays to steady state without any oscillations in shortest possible time.
- Output response is critically damped.
- Closed loop poles will be purely real, equal and negative.

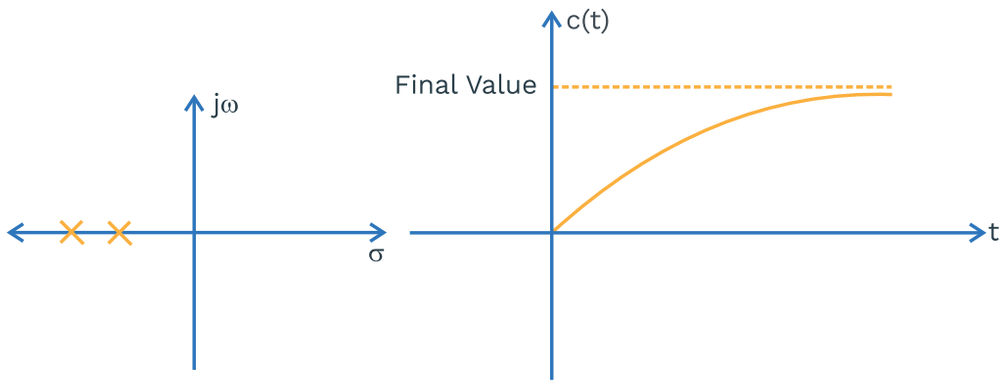
The closed loop poles and corresponding response is given as-



Figure–4.14.

Case – 4: $\xi > 1$

- Output response is overdamped.
- Transients in the system exponentially decays to steady state without any oscillations.
- Closed loop poles will be negative, real and unequal.
- Location of poles & corresponding response can be drawn as –



Figure–4.15.

Characteristic of under damped system

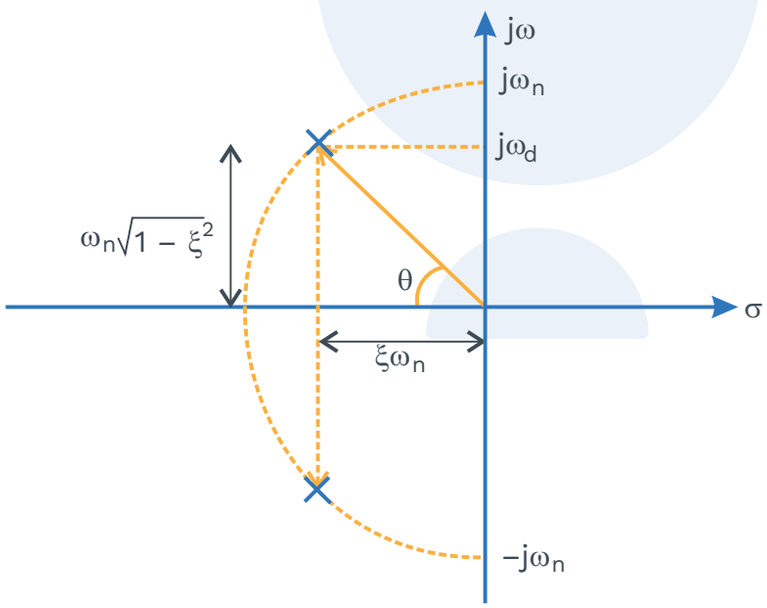
Standard form of second order system is $\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\xi\omega_n s + \omega_n^2}$. The characteristic equation of system is given by –

$$s^2 + 2\xi\omega_n s + \omega_n^2 = 0$$

$$s = -\xi\omega_n \pm j\omega_n\sqrt{1-\xi^2}$$

$$s = -\alpha \pm j\omega_d$$

If we plot closed loop poles on s-plot then, we get:



Figure–4.16.

From the figure,

1. $\sin\theta = \sqrt{1-\xi^2}$

2. $\cos\theta = \frac{\xi\omega_n}{\omega_n} = \xi$

   $\cos\theta = \xi$

   $\Rightarrow \theta = \cos^{-1}\xi$

Previous Year's Questions

The transfer function of a system is given as $\frac{100}{s^2 + 20s + 100}$, the system is

(a) An overdamped system
(b) An underdamped system
(c) A critically damped system
(d) An unstable system

Ans. (c)

3. $\tan\theta = \frac{\sqrt{1-\xi^2}}{\xi}$

4. Damping factor (α)

$$\alpha = \xi\omega_n$$

$$\text{Damping ratio} = \frac{\text{actual damping}}{\text{critical damping}}$$

$$\frac{\xi\omega_n}{\omega_n} = \xi$$

Rack your Brain

What is the difference between damping ratio and damping factor?

Example – 5.

Which of the following system produce undamped oscillation

(a) $\frac{1}{s^2+6s+25}$ (b) $\frac{1}{s^2+4}$

(c) $\frac{1}{s^2+2s+1}$ (d) $\frac{1}{s(s+4)^2}$

Solution:

For an undamped system, $\xi = 0$. Hence, option (b) satisfies.

$$2\xi\omega_n = 0$$

$$\xi = 0$$

Example – 6.

Match the following

	List – I	**List – II**
A.	$s^2 + 5s + 6$	1. Underdamped
B.	$s^2 + 100$	2. Undamped
C.	$s^2 + 2s + 2$	3. Overdamped

	A	**B**	**C**
(a)	2	3	1
(b)	3	2	1
(c)	1	2	3
(d)	3	1	2

Solution: (b)

(A)

$$s^2 + 5s + 6 = 0$$

$$s^2 + 2\xi\omega_n + \omega_n^2 = 0$$

On comparing, $\omega_n^2 = 6, \Rightarrow \omega_n = \sqrt{6}$

$$2\xi\omega_n = 5, \quad \xi = \frac{5}{2 \times \sqrt{6}} = 1.02$$

$\xi = 1.02 > 1$, Over damped

(B)

$$s^2 + 100 = 0$$

$\omega_n = 10$, $2\xi\omega_n = 0$

$\xi = 0$ Undamped

(C)

$$s^2 + 2s + 2 = 0$$

$\omega_n = \sqrt{2}$, $2\xi\omega_n = 2$

$$\xi = \frac{2}{2 \times \sqrt{2}} = \frac{1}{\sqrt{2}} = 0.707$$

$\xi = 0.707 < 1$, Under damped

Step response of second order under damped system –

Transfer function of second order system is

$$\frac{C(s)}{R(s)} = \frac{\omega_n{}^2}{s^2 + 2\xi\omega_n s + \omega_n^2}$$

If input is unit step function,

$$r(t) = u(t), \; R(s) = \frac{1}{s}$$

For step input, output response is given by-

$$C(s) = R(s) \; \frac{\omega_n^2}{s^2 + 2\xi\omega_n s + \omega_n^2}$$

$$C(s) = \frac{1}{s} . \frac{\omega_n^2}{s^2 + 2\xi\omega_n . s + \omega_n^2}$$

By using partial fractions, we can write the above equation as-

$$C(s) = \frac{1}{s} - \frac{s + 2\xi\omega_n}{[(s + \xi\omega_n)^2 + \omega_n^2(1 - \xi^2)]}$$

But

$$\omega_d = \omega_n \sqrt{1 - \xi^2}$$

$$C(s) = \frac{1}{s} - \frac{s + 2\xi\omega_n}{(s + \xi\omega_n)^2 + \omega_d^2}$$

The above equation can be rewritten as-

$$C(s) = \frac{1}{s} - \frac{s + \xi\omega_n}{(s + \xi\omega_n)^2 + \omega_d^2} - \frac{\xi\omega_n}{\omega_d} \cdot \frac{\omega_d}{(s + \xi\omega_n)^2 + \omega_d^2}$$

Taking inverse Laplace transform on both side-

$$\mathcal{L}^{-1}[C(s)] = \mathcal{L}^{-1}\left[\frac{1}{s} - \frac{s + \xi\omega_n}{(s + \xi\omega_n)^2 + \omega_d^2} - \frac{\xi\omega_n}{\omega_d} \cdot \frac{\omega_d}{(s + \xi\omega_n)^2 + \omega_d^2}\right]$$

$$c(t) = 1 - e^{-\xi\omega_n t} \cdot \cos\omega_d t - \frac{\xi\omega_n}{\omega_d} \cdot e^{-\xi\omega_n t} \sin\omega_d t$$

Since,

$$\omega_d = \omega_n\sqrt{1 - \xi^2}$$

$$\therefore c(t) = 1 - \frac{e^{-\xi\omega_n t}}{\sqrt{1 - \xi^2}}\left[\sqrt{(1 - \xi^2)}\cos\omega_d t + \xi\sin\omega_d t\right]$$

Using $(A \sin \omega t + B \cos \omega t) = \sqrt{A^2 + B^2}\left[\sin\left(\omega t + \tan^{-1}\left(\frac{B}{A}\right)\right)\right]$

We get,

$$c(t) = 1 - \frac{e^{-\xi\omega_n t}}{\sqrt{1 - \xi^2}} \sin\left[\omega_d t + \tan^{-1}\left(\frac{\sqrt{1 - \xi^2}}{\xi}\right)\right]$$

$$c(t) = 1 - \frac{e^{-\xi\omega_n t}}{\sqrt{1 - \xi^2}}\sin[\omega_d t + \theta]$$

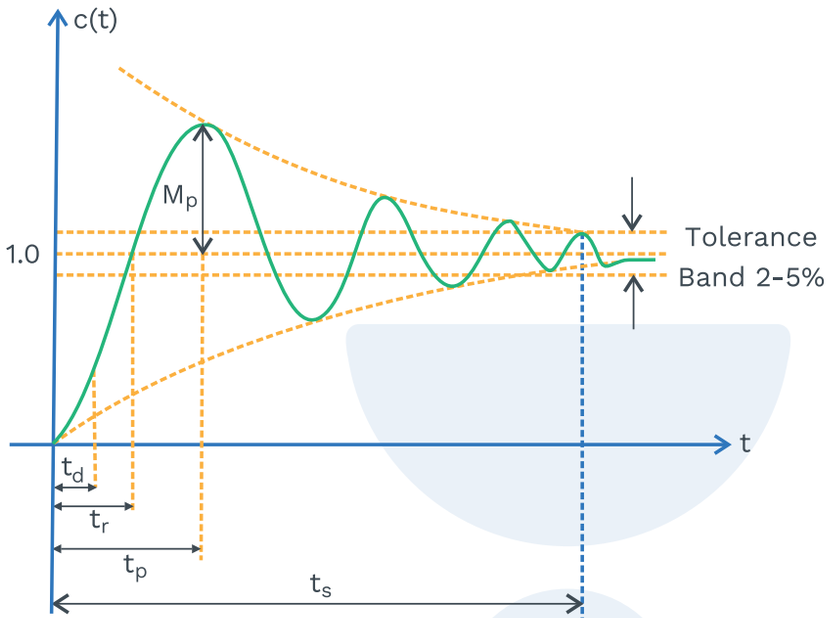
Where,

$$\theta = \cos^{-1}\xi$$

$$\omega_d = \omega_n\sqrt{1 - \xi^2}$$

Time response specifications of second order system-

- These specification refers to the performance indices of the step response of a system.



Figure–4.17.

Delay time (t_d)

Time required for the response to reach 50% of the final value at first instance.

$$t_d = \frac{1 + 0.7\xi}{\omega_n}$$

Rise time (t_r)

Time required for the response to rise from 10% to 90% of the final value for critical damped system and 0 to 100% of the final value for under damped system at first instance.

At rise time (t_r), the time response reaches 100% of the desired value i.e. c(t) = 1

$$c(t) = 1 - \frac{e^{-\xi\omega_n t}}{\sqrt{1-\xi^2}} \sin(\omega_d t + \theta)$$

$$1 = 1 - \frac{e^{-\xi\omega_n t_r}}{\sqrt{1-\xi^2}} \sin(\omega_d t_r + \theta)$$

$$\frac{e^{-\xi\omega_n t_r}}{\sqrt{1-\xi^2}}\sin(\omega_d t_r + \theta) = 0$$

$$\sin(\omega_d t_r + \theta) = 0 = \sin\pi$$

$$\omega_d t_r + \theta = \pi$$

$$\omega_d t_r = \pi - \theta$$

$$t_r = \frac{\pi - \theta}{\omega_d} = \frac{\pi - \cos^{-1}\xi}{\omega_n\sqrt{1-\xi^2}}$$

Where, $\theta = \cos^{-1}\xi$

Peak time (t_p)

- Peak time (t_p) is the time taken by the step response to reach the peak value.
- At peak, the time derivative of response is zero.

$$\therefore \left.\frac{dc(t)}{dt}\right|_{t=t_p} = 0$$

$$\frac{d}{dt}\left[1 - \frac{e^{-\xi\omega_n t}}{\sqrt{1-\xi^2}}.\sin(\omega_d t + \theta)\right] = 0$$

$$0 - \left[\frac{e^{-\xi\omega_n t_p}}{\sqrt{1-\xi^2}}.\omega_d\cos(\omega_d t_p + \theta) - \xi\omega_n\frac{e^{-\xi\omega_n t_p}}{\sqrt{1-\xi^2}}.\sin(\omega_d t_p + \theta)\right] = 0$$

$$\omega_d\cos(\omega_d t_p + \theta) = \xi\omega_n\sin(\omega_d t_p + \theta)$$

$$\tan(\omega_d t_p + \theta) = \frac{\omega_d}{\xi\omega_n}$$

$$\tan(\omega_d t_p + \theta) = \frac{\omega_n\sqrt{1-\xi^2}}{\xi\omega_n}$$

$$\tan(\omega_d t_p + \theta) = \frac{\sqrt{1-\xi^2}}{\xi}$$

$$\Rightarrow \quad \tan(\omega_d t_p + \theta) = \tan\theta$$

$$\because \quad \tan(n\pi + \theta) = \tan\theta$$

$$\therefore \quad \omega_d t_p = n\pi$$

$$t_p = \frac{n\pi}{\omega_d} \text{ seconds}$$

Where, n = odd term (1, 3, 5.....) for overshoot
n = even term (2, 4, 6.....) for under shoot

First peak overshoot (n = 1) = $\frac{\pi}{\omega_d}$

First peak undershoot (n = 2) = $\frac{2\pi}{\omega_d}$

Peak overshoot (M_p)

- It is the normalised difference between the peak value of time response and the steady state value.
- The first peak in the response is maximum peak.

Maximum peak overshoot = $C_{max} - C_{ss}$

$$C_{max} = c(t)\big|_{t=t_p}$$

$$C_{max} = c(t_p) = \left[1 - \frac{e^{-\xi\omega_n t_p}}{\sqrt{1-\xi^2}} \sin(\omega_d t_p + \theta)\right]$$

$$\because t_p = \frac{\pi}{\omega_d}, n = 1 \text{ for first overshoot}$$

$$C_{max} = \left[1 - \frac{e^{-\xi\omega_n t_p}}{\sqrt{1-\xi^2}} \sin(\omega_d \times \frac{\pi}{\omega_d} + \theta)\right]$$

$$= \left[1 - \frac{e^{-\xi\omega_n t_p}}{\sqrt{1-\xi^2}} (-\sin\theta)\right]$$

$$= \left[1 + \frac{e^{-\xi\omega_n t_p}}{\sqrt{1-\xi^2}} \sqrt{1-\xi^2}\right]$$

$$= \left[1 + e^{-\xi\omega_n t_p}\right]$$

$$= 1 + e^{-\xi\omega_n \times \frac{\pi}{\omega_d}} = 1 + e^{-\xi\omega_n \times \frac{\pi}{\omega_n\sqrt{1-\xi^2}}}$$

$$C_{max} = 1 + e^{-\pi\xi/\sqrt{1-\xi^2}} \quad \text{..... (i)}$$

And $C_{ss} = 1$

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Time difference between successive overshoot or under shoot = $\frac{2\pi}{\omega_d}$ (time period of oscillations)

So, maximum peak overshoot (M_p) = $C_{max} - C_{ss}$

$$M_p = 1 + e^{-\pi\xi/\sqrt{1-\xi^2}} - 1$$

$$M_p = e^{-\pi\xi/\sqrt{1-\xi^2}} \quad \text{..... (ii)}$$

$$\% \text{ maximum peak overshoot} = \frac{C_{max} - C_{ss}}{C_{ss}} \times 100\%$$

$$\%M_p = e^{-\pi\xi/\sqrt{1-\xi^2}} \times 100\%$$

Where, n = 1, 3, 5 ..... for overshoots
n = 2, 4, 6 ..... for undershoots

$$\text{Absolute peak overshoot} = \left(e^{-\pi\xi/\sqrt{1-\xi^2}}\right) C_{ss}$$

Settling time

- Time required for the response to reach and stay within a specified tolerance band (take 2%) of its final value or steady state value.

For a second order under damped control system,
For 2% tolerance band

$$t_s = 4\tau$$

$$t_s = \frac{4}{\xi\omega_n}$$

For 5% tolerance band

$$t_s = 3\tau = \frac{3}{\xi\omega_n}$$

Number of cycle

Number of cycle before reaching the steady state is-

$$\omega_d = 2\pi f_d$$

$$f_d = \frac{\omega_d}{2\pi}$$

No. of cycle = $t_s \times f_d$
For 2% tolerance band –
No. of cycle = $4T \times f_d$

$$\text{No. of cycle} = \frac{4f_d}{\xi\omega_n}$$

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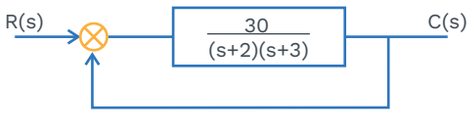
- If instead of unit step (u(t)), we apply step signal (Au(t)), then percentage overshoot remains same, but absolute overshoot changes.
- Maximum overshoot is used to measure the relative stability of control system.
- Maximum overshoot increases then relative stability decreases.

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The equation of the rise time, peak time, maximum overshoot and setting time are valid only for standard second order system.

Example – 7.

For the given closed loop system, find damping ratio undamped natural frequency, damped angular frequency, first peak overshoot time, first peak undershoot time, maximum % overshoot, number of cycle completed before reaching the steady state condition.



Solution:

Overall transfer function is given by

$$\frac{C(s)}{R(s)} = \frac{\dfrac{30}{(s+2)(s+3)}}{1+\dfrac{30}{(s+2)(s+3)}\cdot 1}$$

$$\frac{C(s)}{R(s)} = \frac{30}{s^2+5s+36}$$

Characteristic equation is –

$$s^2+5s+36=0$$

Comparing with $s^2+2\xi\omega_n s+\omega_n^2=0$, we get-

$$\omega_n^2 = 36 \Rightarrow \omega_n = 6$$

$$2\xi\omega_n = 5$$

$$\xi = \frac{5}{2\times 6} = 0.41$$

$$\omega_d = \omega_n\sqrt{1-\xi^2} = 6\sqrt{1-(0.4)^2}$$

$$\omega_d = 5.47 \text{ rad / sec}$$

t_{po} (first peak overshoot) $= \dfrac{\pi}{\omega_d} = \dfrac{\pi}{5.47}$ sec

t_{pu} (first peak undershoot) $= \dfrac{2\pi}{5.47} = \dfrac{\pi}{2.73}$ sec

$$M_p = e^{-\pi\xi/\sqrt{1-\xi^2}}\times 100 = e^{-0.4\pi/\sqrt{1-(0.4)^2}}$$

$$= e^{-1.37}\times 100 = 25.4\%$$

Number of cycles completed before reaching steady state

$$= \frac{\omega_d}{2\pi}\times\frac{4}{\xi\omega_n} = \frac{5.47}{2\pi}\times\frac{4}{0.41\times 6}$$

= 1.41 cycles

Previous Year's Questions

Which of the following has maximum peak overshoot due to a unit step input?

(a) $\dfrac{100}{s^2+10s+100}$

(b) $\dfrac{100}{s^2+15s+100}$

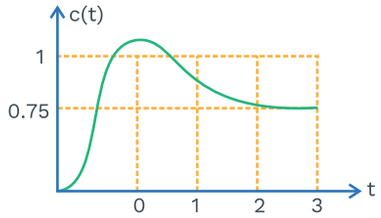
(c) $\dfrac{100}{s^2+5s+100}$

(d) $\dfrac{100}{s^2+20s+100}$

Ans. (d)

Example – 8.

The unit step response of a unity feedback system with open loop transfer function $G(s) = \frac{K}{(s+1)(s+2)}$ is shown in figure. The value of K is:



Solution:

From the graph

$$e_{ss} = 1 - 0.75 = 0.25$$

$$\text{Error} = E(s) = \frac{R(s)}{1 + G(s)H(s)}$$

$$R(s) = \frac{1}{s},\ H(s) = 1$$

$$G(s) = \frac{K}{(s+1)(s+2)}$$

$$e_{ss} = \lim_{s \to 0} s.E(s)$$

$$= \lim_{s \to 0} \frac{s.\frac{1}{s}}{1 + \frac{K}{(s+1)(s+2)}} = \frac{1}{1 + \frac{K}{2}}$$

$$0.25 = \frac{1}{1 + \frac{K}{2}}$$

$$\Rightarrow 1 + \frac{K}{2} = 4 \quad \Rightarrow K = 6$$

Example – 9.

The block diagram of a unity feedback control system has

$$G(s) = \frac{K}{s(s+8)}$$

Determine the value of K and peak overshoot to unit step input of damping ratio 0.7.

Solution:

For the given system-

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1+G(s)} = \frac{K}{s^2+8s+K}$$

$$s^2 + 8s + K = 0$$

$$\omega_n = \sqrt{K}\ ,\ 2\xi\omega_n = 8$$

$$\xi = \frac{8}{2\sqrt{K}} \Rightarrow \sqrt{K} = \frac{4}{\xi} \Rightarrow K = \frac{16}{\xi^2} = \frac{16}{0.7^2}$$

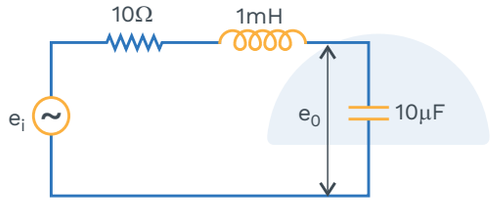
$$K = 32.65$$

$$M_p = e^{-\pi\xi/\sqrt{1-\xi^2}} = e^{-\pi 0.7/\sqrt{1-0.7^2}}$$

$$M_p = 0.046 = 4.6\%$$

Example – 10.

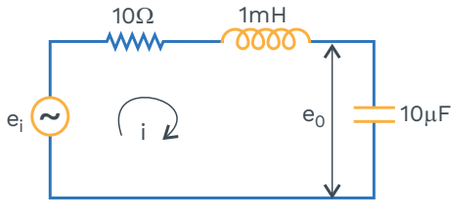
RLC circuit shown in figure



Determine overshoot in the output e_0 for a step input.

Solution:

Apply KVL- $e_i = Ri + L.\frac{di}{dt} + \frac{1}{C}\int i\,dt$



Taking Laplace transform –

$$E_i(s) = \left(R + sL + \frac{1}{sC}\right)I(s)$$

$$I(s) = \frac{E_i(s)}{R + sL + \frac{1}{sC}}$$

$$e_0 = \frac{1}{C}\int i.\,dt$$

$$E_0(s) = \frac{1}{sC} I(s) = \frac{1}{sC}\left[\frac{E_i(s)}{R + sL + \frac{1}{sC}}\right]$$

$$\frac{E_0(s)}{E_i(s)} = \frac{1}{LC\left(s^2 + \frac{R}{L}s + \frac{1}{LC}\right)}$$

Characteristic equation:

$$s^2 + \frac{R}{L}s + \frac{1}{LC} = 0$$

Comparing with, $s^2 + 2\xi\omega_n s + \omega_n^2 = 0$

$$\omega_n = \frac{1}{\sqrt{LC}}; 2\xi\omega_n = \frac{R}{L}$$

$$\xi = \frac{R}{2}\sqrt{\frac{C}{L}}$$

$$\xi = \frac{10}{2} \times \sqrt{\frac{10 \times 10^{-6}}{1 \times 10^{-3}}} = 0.5$$

$$\% M_P = e^{-\pi\xi/\sqrt{1-\xi^2}} \times 100$$

$$= e^{-\pi 0.5/\sqrt{1-0.5^2}} \times 100$$

$$= 0.163 \times 100$$

$$= 16.3\%$$

Step response of second order undamped system–

Standard form of second order transfer function is

$$T(s) = \frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\xi\omega_n s + \omega_n^2}$$

For undamped system, $\xi = 0$

$$T(s) = \frac{\omega_n^2}{s^2 + \omega_n^2}$$

Previous Year's Questions

Transfer function of a plant is:

$$T(s) = \frac{5}{(s+5)(s^2+s+1)}$$

The second order approximation of T(s) using dominant pole concept is:

(a) $\frac{1}{(s+5)(s+1)}$

(b) $\frac{5}{(s+5)(s+1)}$

(c) $\frac{5}{s^2+s+1}$

(d) $\frac{1}{s^2+s+1}$

Ans. (d)

If input is step input

$$r(t) = u(t), R(s) = \frac{1}{s}$$

$$C(s) = \frac{\omega_n^2}{s(s^2 + \omega_n^2)}$$

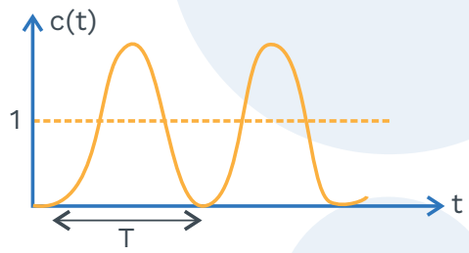
By using partial fraction-

$$C(s) = \frac{1}{s} - \frac{s}{s^2 + \omega_n^2}$$

By taking inverse Laplace transform

$$c(t) = [1 - \cos \omega_n t]u(t)$$

If we plot the response graphically, then –



Dominant pole

- Closed loop poles that have dominant effects on the transient response are called dominant pole or the pole which is nearer to the origin is called as dominant pole.
- We know,

$$\text{Time constant} = \frac{-1}{\text{real part of the pole}}$$

The pole which is far away from origin has reduced τ as compared to pole near origin, then the pole may be regarded as insignificant poles as far as the transient response is concerned.

For example, consider a system having two closed loop poles:

$$G(s) = \frac{K}{(s + p_1)(s + p_2)}$$

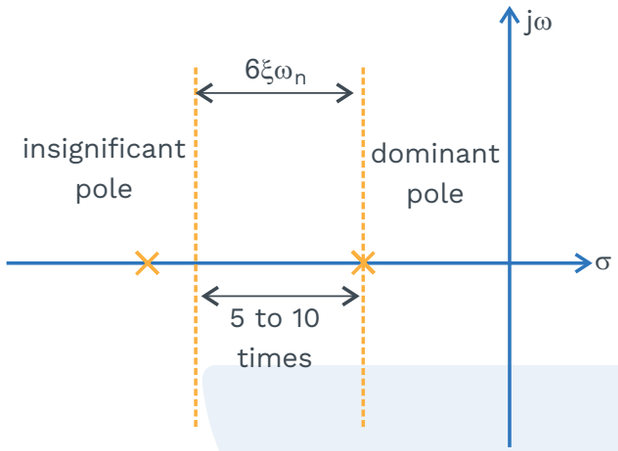
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Why is a dominant pole required in control system?

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The value of DC gain of system should be same while applying the concept of dominant pole of transfer function.

If $\left|\frac{p_2}{p_1}\right| \geq 5$, then p_2 can be ignored, which can be shown as graphically:



- If magnitude of the real part of a pole is at least 5 to 10 times of that of a dominant pole. Then, insignificant pole can be neglected.

Example - 11

A closed loop transfer function is given as:

$$T(s) = \frac{100}{(s+2)(s+12)}$$

Calculate the step response by using dominant pole analysis.

Solution:

$$T(s) = \frac{100}{(s+2)(s+12)}$$

Poles are $p_1 = -2, p_2 = -12$

$$\left|\frac{p_2}{p_1}\right| = \frac{12}{2} = 6 \geq 5$$

then, $p_2 = -12$ can be ignored. Thus, the transfer function will be:

$$T(s) = \frac{100}{12(s+2)} = \frac{25}{3(s+2)}$$

$$C(s) = R(s)\frac{25}{3(s+2)}$$

$$C(s) = \frac{1}{s}\frac{25}{3(s+2)}$$

By using partial fraction-

$$C(s) = \frac{25}{6s} - \frac{25}{6(s+2)}$$

$$\Rightarrow\ c(t) = \frac{25}{6}\left[1 - e^{-2t}\right]u(t)$$

Response of higher order system

- Any higher order system can be factored as a product of first order terms and one quadratic term.
- For example, a third order system can be considered as a combination of first order and second order underdamped system. Thus, the transfer function of higher order system can be written as:

$$\frac{C(s)}{R(s)} = \frac{K}{(s+p_1)(s+p_2).....(s^2 + 2\xi\omega_n s + \omega_n^2)}$$

- The time response analysis of higher order system is obtained by approximating them to second order system using dominant pole concept.

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In time response analysis of higher order system, first convert the transfer function in time constant form then apply the concept of dominant pole.

Example 12:

Calculate the approximate transfer function by using dominant pole analysis.

$$\frac{C(s)}{R(s)} = \frac{288}{(s+2)(s+5)(s+12)(s+24)}$$

Solution:

$$\frac{C(s)}{R(s)} = \frac{288}{(s+2)(s+5)(s+12)(s+24)}$$

Poles are, $p_1 = -2$ $\quad p_3 = -12$

$p_2 = -5$ $\quad p_4 = -24$

So that, we can ignore the pole p_3 and p_4 to bring order to second order system. Thus:

$$\frac{C(s)}{R(s)} = \frac{288}{(s+2)(s+5)\times 12\times 24}$$

$$\frac{C(s)}{R(s)} = \frac{1}{(s+2)(s+5)}$$

Steady state error

- It is an index of accuracy of a control system, therefore it should be minimum.

Consider a closed loop control system represented by a block diagram is-



Where,
R(s) = reference input
C(s) = output
G(s) = forward path gain
H(s) = feedback gain
E(s) = error signal
The error E(s) and the input R(s) are related as follows:

$$\frac{E(s)}{R(s)} = \frac{1}{1+G(s)H(s)}$$

$$E(s) = R(s)\frac{1}{1+G(s)H(s)}$$

Apply final value theorem, steady state error is:

$$e_{ss} = \lim_{t\to\infty} e(t)$$

$$e_{ss} = \lim_{s\to 0} sE(s) = \lim_{s\to 0} s\frac{R(s)}{1+G(s)H(s)}$$

$$e_{ss} = \lim_{s\to 0} s\frac{R(s)}{1+G(s)H(s)}$$

Steady state error for different type of input–
The expression for steady state errors for various types of standard input signals are derived below-

1. Unit step input
For unit step input –

$$r(t) = u(t)$$

or $R(s) = \frac{1}{s}$

Definition

Steady state error is the error between the actual output and the desired output at $t \to \infty$.

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- Steady state error depends on type of the reference input and type of the system.
- Steady state error is calculated only for closed loop stable system.

So, $e_{ss} = \lim_{s\to 0} \frac{s\ R(s)}{1 + G(s)\,H(s)}$

$$e_{ss} = \lim_{s\to 0} s. \frac{\frac{1}{s}}{1 + \lim_{s\to 0} G(s)\,H(s)} = \frac{1}{1 + \lim_{s\to 0} G(s)}$$

$$e_{ss} = \frac{1}{1 + K_p}$$

Where, $K_p = \lim_{s\to 0} G(s)$ is called static positional error coefficient.

2. Unit ramp input

Input is $r(t) = t\ u(t)$

$$R(s) = \frac{1}{s^2}$$

$$e_{ss} = \lim_{s\to 0}\ s. \frac{R(s)}{1 + G(s)H(s)}$$

$$e_{ss} = \lim_{s\to 0} \frac{s. \frac{1}{s^2}}{1 + G(s)H(s)}$$

$$e_{ss} = \lim_{s\to 0} \frac{1}{s + s\ G(s)\ H(s)}$$

$$= \lim_{s\to 0} \frac{1}{s\,G(s)\ H(s)}$$

$$= \frac{1}{\lim_{s\to 0} s\,G(s)\ H(s)} = \frac{1}{K_v}$$

$$e_{ss} = \frac{1}{K_v}$$

Where, K_v = static velocity error coefficient

and $K_v = \lim_{s\to 0} s\,G(s)\ H(s)$

3. Unit parabolic input –

Input is $r(t) = t^2 u(t)$

or $R(s) = \frac{1}{s^3}$

$$e_{ss} = \lim_{s\to 0} \frac{s.\ R(s)}{1 + G(s)\ H(s)}$$

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Velocity error is not error in the velocity but it is error in position due to ramp input.

$$= \lim_{s\to 0} \frac{s.\frac{1}{s^3}}{1 + G(s)\ H(s)}$$

$$= \lim_{s\to 0} \frac{1}{s^2 + s^2.G(s)\ H(s)}$$

$$= \frac{1}{\lim_{s\to 0} s^2\ G(s)\ H(s)}$$

$$e_{ss} = \frac{1}{K_a}$$

Where, K_a = static acceleration error coefficient

And $K_a = \lim_{s\to 0} s^2\ G(s)\ H(s)$

Type of a system

Consider the following pole – zero form of open loop transfer function of a system.

$$G(s) = \frac{K(s+z_1)(s+z_2).....}{s^n(s+p_1)(s+p_2).....}$$

Term s^n in the denominator denotes the number of poles (n) at origin.

- System with n poles at origin is called as type-n system.
- As $s \to 0$, s^n term dominates in determining the steady state error.

Steady state error for different types of system–

The steady state error depends on the type of open loop transfer function of a closed loop system.

1. (a) Type – '0' system with unit step input

For type '0' system, the open loop transfer function is given by-

$$G(s)H(s) = \frac{K(1+sz_1)(1+sz_2).....}{(1+sp_1)(1+sp_2).....}$$

$$K_p = \lim_{s\to 0} G(s).H(s)$$

$$G(s)H(s) = \lim_{s\to 0} \frac{K(1+sz_1)(1+sz_2).....}{(1+sp_1)(1+sp_2).....}$$

Rack your Brain

What is the drawback of static coefficients?

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The error constant K_p, K_v and K_a describe the ability of a system to eliminate steady state errors.

$$K_p = K$$

$$e_{ss} = \frac{1}{1+K_p} = \frac{1}{1+K}$$

(b) Type '0' system with unit ramp input

Transfer function is

$$G(s)H(s) = \frac{K(1+sz_1)(1+sz_2).....}{(1+sp_1)(1+sp_2).....}$$

$$K_v = \lim_{s\to 0} \; s.\; G(s)\;\; H(s)$$

$$= \lim_{s\to 0} s. \frac{K(1+sz_1)(1+sz_2).....}{(1+sp_1)(1+sp_2).....}$$

$$K_v = 0$$

$$\therefore \; e_{ss} = \frac{1}{K_v} = \frac{1}{0} = \infty$$

$$e_{ss} = \infty$$

(c) Type '0' system with unit parabolic input

$$G(s)\;H(s) = \frac{K(1+sz_1)(1+sz_2).....}{(1+sp_1)(1+sp_2).....}$$

$$K_a = \lim_{s\to 0} s^2\; G(s)\;H(s)$$

$$= \lim_{s\to 0} s^2 . \frac{K(1+sz_1)(1+sz_2).....}{(1+sp_1)(1+sp_2).....}$$

$$K_a = 0$$

$$\therefore \; e_{ss} = \frac{1}{K_a} = \frac{1}{0} = \infty$$

$$e_{ss} = \infty$$

2. (a) Type '1' system with unit step input

For type '1' system the open loop transfer function is given by-

$$G(s)H(s) = \frac{K(1+sz_1)(1+sz_2).....}{s(1+sp_1)(1+sp_2).....}$$

$$K_p = \lim_{s\to 0} G(s)H(s)$$

$$= \lim_{s\to 0} \frac{K(1+sz_1)(1+sz_2).....}{s(1+sp_1)(1+sp_2).....}$$

$$K_p = \infty$$

$$\therefore\ e_{ss} = \frac{1}{1+K_p} = \frac{1}{1+\infty} = 0$$

$$e_{ss} = 0$$

(b) Type '1' system with unit ramp input

$$G(s)H(s) = \frac{K(1+sz_1)(1+sz_2).....}{s(1+sp_1)(1+sp_2).....}$$

$$K_v = \lim_{s\to 0} s.\,G(s)H(s)$$

$$= \lim_{s\to 0} s. = \frac{K(1+sz_1)(1+sz_2).....}{s(1+sp_1)(1+sp_2).....}$$

$$K_v = K$$

$$\therefore\ e_{ss} = \frac{1}{K_v} = \frac{1}{K}$$

(c) Type '1' system with unit parabolic input

$$G(s)\ H(s) = \frac{K(1+sz_1)(1+sz_2)}{s(1+sp_1)(1+sp_2)}$$

$$K_a = \lim_{s\to 0} s^2 G(s)\ H(s)$$

$$= \lim_{s\to 0} s^2\ \frac{K(1+sz_1)(1+sz_2)}{s(1+sp_1)(1+sp_2)}$$

$$K_a = 0$$

$$e_{ss} = \frac{1}{K_a} = \frac{1}{0} = \infty$$

$$e_{ss} = \infty$$

3. (a) Type '2' system with unit step input

$$G(s)\ H(s) = \frac{K(1+sz_1)(1+sz_2).....}{s^2(1+sp_1)(1+sp_2).....}$$

$$K_p = \lim_{s\to 0} G(s)\ H(s)$$

$$= \lim_{s\to 0} .\ \frac{K(1+sz_1)(1+sz_2).....}{s^2(1+sp_1)(1+sp_2).....}$$

$$K_p = \infty$$

$$e_{ss} = \frac{1}{1+K_p} = \frac{1}{1+\infty} = \frac{1}{\infty} = 0$$

$$e_{ss} = 0$$

(b) Type '2' system with unit ramp input

$$G(s)\ H(s) = \frac{K(1+sz_1)(1+sz_2).....}{s^2(1+sp_1)(1+sp_2).....}$$

$$K_v = \lim_{s\to 0} s\ G(s)\ H(s)$$

$$= \lim_{s\to 0} s \frac{K(1+sz_1)(1+sz_2).....}{s^2(1+sp_1)(1+sp_2).....}$$

$$K_v = \infty$$

$$e_{ss} = \frac{1}{K_v} = \frac{1}{\infty}$$

$$e_{ss} = 0$$

(c) Type '2' system with unit parabolic input

$$G(s)\ H(s) = \frac{K(1+sz_1)(1+sz_2).....}{s^2(1+sp_1)(1+sp_2).....}$$

$$K_a = \lim_{s\to 0} .s^2\ G(s)\ H(s)$$

$$= \lim_{s\to 0} s^2 . \frac{K(1+sz_1)(1+sz_2).....}{s^2(1+sp_1)(1+sp_2).....}$$

$$K_a = K$$

$$\therefore\ e_{ss} = \frac{1}{K_a} = \frac{1}{K}$$

$$e_{ss} = \frac{1}{K}$$

The steady state error as per the type of an open loop transfer function with specified input is shown in the table below.

Input	Type-0		Type-1		Type-2	
	Static error coefficient	e_{ss}	Static error coefficient	e_{ss}	Static error coefficient	e_{ss}
Unit step	$K_p = K$	$\frac{1}{1+K}$	$K_p = \infty$	0	$K_p = \infty$	0
Unit ramp	$K_v = 0$	∞	$K_v = K$	$\frac{1}{K}$	$K_v = \infty$	0
Unit parabolic	$K_a = 0$	∞	$K_a = 0$	∞	$K_a = K$	$\frac{1}{K}$

- For type '1' system, parabolic input is not acceptable.
- For type '2' system, all three inputs are acceptable.

Example-13.

The OLTF of an unity feedback system is $\frac{(s-5)}{s^2+7s+1}$. The steady state error to a step unit input is-

Solution:

Characteristic equation is 1 + G(s) = 0

$$G(s) = \frac{(s-5)}{s^2+7s+1}$$

$$1+\frac{(s-5)}{s^2+7s+1} = 0$$

$$s^2 + 8s - 4 = 0$$

By Routh Hurwitz Criteria, system is unstable.

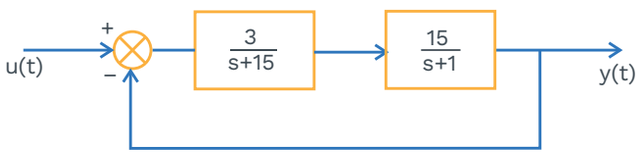
$\therefore$ Error is unbounded.

Gray Matter Alert!!!

- Steady state error is inversely proportional to the forward path gain K.
- For type '0' system, ramp and parabolic inputs are not acceptable.

Example-14

The block diagram shown in figure given a unity feedback closed loop control system. The steady state error in response of the above system to unit step is:



Solution:

$$G(s) = \frac{3}{s+15} \times \frac{15}{s+1}$$

$$G(s) = \frac{45}{(s+1)(s+15)}, \quad H(s) = 1$$

$$G(s)\,H(s) = \frac{45}{(s+1)(s+15)}$$

Type of the system is zero

$$\therefore \quad e_{ss} = \frac{1}{1+K_p}$$

$$K_p = \lim_{s\to 0} G(s)H(s) = \lim_{s\to 0} \frac{45}{(s+1)(s+15)}$$

$$K_p = 3$$

$$e_{ss} = \frac{1}{1+3} = \frac{1}{4} = 0.25$$

Dynamic error coefficients –

- The dynamic (generalised) error coefficients to be described below gives the steady state error performance of a feedback control system to any arbitrary input function.

The error transfer function of control system is,

$$\frac{E(s)}{R(s)} = \frac{1}{1+G(s)H(s)}$$

$$\text{Let } F(s) = \frac{1}{1+G(s)H(s)}$$

$$\therefore\ E(s) = R(s)\ .\ F(s)$$

Using convolution theorem the inverse Laplace transform of E(s) is:

$$e(t) = f(\tau) * r(t)$$

$$= \int_{-\infty}^{t} f(\tau) * r(t-\tau)d\tau$$

Expanding r (t – τ) using Taylor series –

$$r(t-\tau) = r(t) - \tau\, r'(t) + \frac{\tau^2}{2!}r''(t) - \frac{\tau^3}{3!}r'''(t) + \ldots.$$

$$e(t) = r(t)\int_0^{\infty} f(\tau).d\tau - r'(t)\int_0^{\infty} \tau f(\tau).d\tau + r''(t)\int_0^{\infty} \frac{\tau^2}{2!}f(\tau).d\tau$$

Defining dynamic error constants –

$$C_0 = \int_0^{\infty} f(\tau)d\tau$$

$$C_1 = \int_0^{\infty} \tau f(\tau)d\tau$$

$$C_2 = \int_0^{\infty} \frac{\tau^2}{2!}f(\tau)d\tau$$

$$\Rightarrow e(t) = C_0 r(t) - C_1\, r'(t) + C_2\, r''(t) - C_3\, r'''(t) + \ldots \frac{C_n}{n!}r^n(t)$$

Previous Year's Questions

A unity feedback system has open loop transfer function G(s). The steady state error is zero for.
(a) Step input and type 1 G(s)
(b) Ramp input and type 1 G(s)
(c) Step input and type 0 G(s)
(d) Ramp input and type 0 G(s)
Ans. (a)

Dynamic error constant –

As per Laplace transform,

$$F(s) = \int_0^\infty f(\tau)e^{-s\tau}.d\tau$$

And $$\lim_{s\to0} F(s) = \lim_{s\to0} \int_0^\infty f(\tau)e^{-s\tau}d\tau$$

$$= \int_0^\infty f(\tau).d\tau = C_0$$

$$\Rightarrow C_0 = \lim_{s\to0} F(s)$$

Similarly, $C_1 = \lim_{s\to0} \frac{dF(s)}{ds}$

$$C_2 = \lim_{s\to0} \frac{d^2F(s)}{ds}$$

Where, $F(s) = \frac{1}{1+G(s).H(s)}$

Rack your Brain

What are the main advantages of dynamic error constant?

Example 14.

An open loop TF $G(s) = \frac{9}{s+1}$ at unity feedback system. Evaluate the first three coefficient error series and steady state error for input $r(t) = \frac{3t^2}{2}$.

Solution:

$$G(s) = \frac{9}{s+1}, \quad H(s) = 1$$

$$r(t) = \frac{3t^2}{2}$$

$$C_0 = \lim_{s\to0} F(s)$$

$$F(s) = \frac{1}{1+G(s)H(s)} = \frac{1}{1+\frac{9}{s+1}} = \frac{s+1}{s+10}$$

$$C_0 = \lim_{s\to0} \frac{s+1}{s+10} = \frac{1}{10} = 0.1$$

$$C_1 = \lim_{s\to0} \frac{dF(s)}{ds} = \lim_{s\to0} \frac{d}{ds}\left(\frac{s+1}{s+10}\right)$$

$$= \lim_{s\to0}\left[\frac{(s+10)-(s+1)}{(s+10)^2}\right] = \frac{9}{100}$$

$$C_1 = 0.09$$

$$C_2 = \lim_{s\to 0}\frac{d^2F(s)}{ds} = \lim_{s\to 0}\frac{d}{ds}\left(\frac{9}{(s+10)^2}\right)$$

$$= \lim_{s\to 0}\left[\frac{d}{ds}9(s+10)^{-2}\right]$$

$$= \lim_{s\to 0}\left[9(-2)(s+10)^{-3}\right] = \frac{-18}{1000}$$

$$C_2 = -0.018$$

Error series is given by –

$$e(t) = C_0 r(t) + C_1 r'(t) + C_2 r''(t)$$

$$r(t) = \frac{3t^2}{2}$$

$$r'(t) = \frac{6t}{2} = 3t$$

$$r''(t) == 3.(1) = 3 \text{ and } r'''(t) = 0$$

$$\therefore\ e(t) = (0.1)\frac{3t^2}{2} + (0.09)3t + (-0.018).3$$

$$e(t) = 0.15t^2 + 0.27t - 0.054$$

$$e_{ss} = \lim_{t\to\infty} e(t)$$

$$= \lim_{t\to\infty}[0.15t^2 + 0.27t - 0.054]$$

$$e_{ss} = \infty$$

PRACTICE QUESTIONS

Q1 **The maximum overshoot and settling time for a control system whose overall transfer function is-**

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\xi\omega_n s + \omega_n^2}$$

are 10% and 4 second respectively, the input being a unit step function. Determine the value of ξ and ω_n in order to satisfy these requirements.

Sol. $\xi = 0.59, \quad \omega_n = 1.69$ rad/sec

Q2 **Determine the unit step time response of a system having closed loop transfer function given below.**

$$\frac{C(s)}{R(s)} = \frac{1}{(s+1)(s^2+1)}$$

Sol. $c(t) = 1 - \frac{e^{-2t}}{2} - \frac{1}{\sqrt{2}}\sin(t + 45^\circ)$

Q3 **Determine the steady state error for a system whose open loop transfer function is given by** $G(s) = \frac{5(s+4)}{s(s+1)(s+5)}$ **and input to the system being**

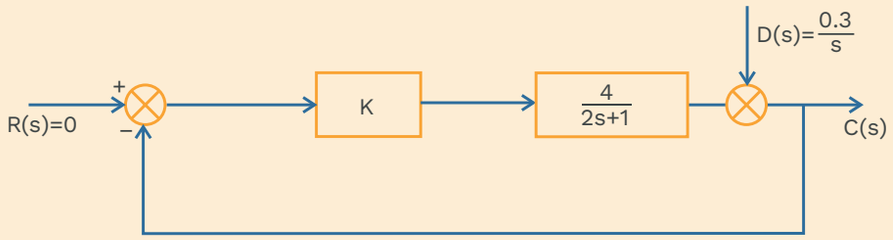
r(t) = 0 for t < 0
r(t) = 2+2t for t ≥ 0

Sol. $e_{ss} = 0.5$

Q4 **The steady state error (e_{ss}) with respect to type**
(a) e_{ss} increases as type increases (b) e_{ss} decreases as type increases
(c) e_{ss} not depends on type (d) e_{ss} depends on type

Sol. **Correct answer is (b).**

Q5 **Consider the system shown below –**



The steady state offset due to D(s) will be zero for which value of K?

Sol. $K = \infty$

Q6 **Time response of a standard second order system to a step input is:**

$$1 - Ke^{-2t}.\sin\left(2\pi t + \phi\right)$$

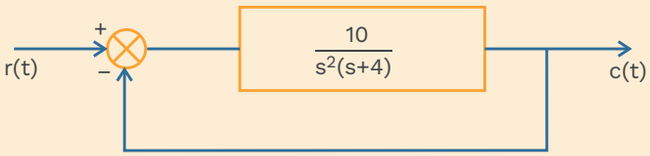
Find number of oscillations if tolerance band is 2%

Sol. **2**

Q7 **If the roots of a second order characteristic equation are given by $s_{1,2} = -3 \pm j2$ then find the value of damping ratio and the damped natural frequency?**

Sol. $\xi = \frac{3}{\sqrt{13}}$, $\omega_d = 2\,\text{rad / sec}$

Q8 **The steady state error resulting from input $r(t) = 2+3t + 4t^2$ for given system is:**



Sol. System is unstable , error is unbounded.

Q9 **The open loop transfer function of different systems is given below. Among these, the system with the highest steady state error to a step input is:**

(a) $\frac{120}{(s+1)(s+4)(s+15)}$

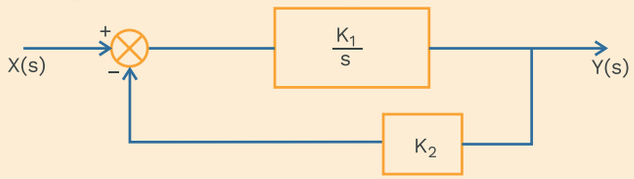
(b) $\frac{60}{(s+0.5)(s+3)(s+5.5)}$

(c) $\frac{80}{(s+1)(s+2)(s+3)}$

(d) $\frac{120}{(s+1)(s+15)}$

Sol. **Correct answer is (a).**

Q10 **In the block diagram shown in the figure below**



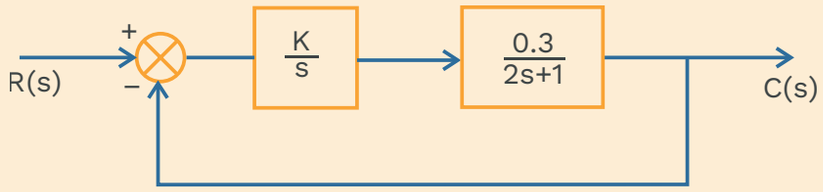
Find the value of K_1, K_2 for which the temperature constant is 0.2 seconds and steady state output is 4 for a step unit.

Sol. $K_1 = 20, \quad K_2 = 0.25$

Q11 **A speed control system is expressed by transfer function $\frac{\omega(s)}{v(s)} = \frac{100}{s+100}$. The steady state output speed of the given system for a unit step voltage input is _____ (rad/sec)**

Sol. 1

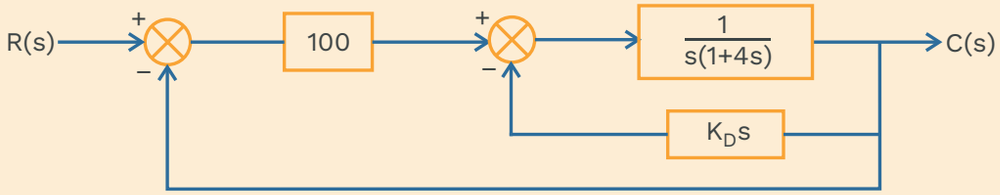
Q12 **For the system shown below,**



Find the value of K that results in critical damping?

Sol. K = 0.417

Q13 **A control system is being represented by the block diagram as shown below. To improve the damping of the system the output rate control is used. If the damping factor of a closed loop system is 0.5, find the value of K_D?**



Sol. $K_D = 19$

Q14 **The block diagram of a unity feedback control system has, $G(s) = \frac{K}{(s+8)}$.**

Find the value of K and peak overshoot to unit step input.

Sol. K = 32.7, $M_p = 0.046$

Q15 **A unity feedback system has an open loop transfer function given as**

$$G(s) = \frac{K}{s(1+Ts)}$$

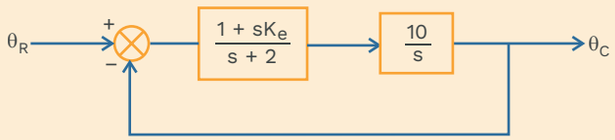
Where K and T are positive constants. Then the factor by which amplifier gain can be reduced so that the peak overshoot of unit-step response of the system is reduced from 75% to 25%.

Sol. 20

Q16 **The response of a servomechanism is given as $c(t) = 1 + 0.2\,e^{-60t} - 1.2\,e^{-10t}$ When the response is subjected to a unit step input.**
Obtain:
(a) The expression for the closed loop transfer function.
(b) The undamped natural frequency and damping ratio of the system.

Sol. (a) $\frac{600}{(s+10)(s+60)}$ (b) 24.5 rad/sec and 1.43

Q17 **Obtain the values of the error rate factor such that the damping ratio is 0.5 and also determine the values of settling time, maximum overshoot for unit ramp input with and without error rate control.**



Sol. $K_e = 0.116$
With feedback: $t_s = 2.53$ sec., $M_p = 16.2\%$
Without feedback: $t_s = 4$ sec., $M_p = 35.2\%$

Q18 **A unity feedback system is characterized by the open loop transfer function is**

$$G(s) = \frac{1}{s(1+0.5s)(1+0.2s)}$$

Determine the steady state errors for unit step, unit ramp and unit acceleration inputs.

Sol. 0, 1, ∞

Q19 **A closed loop control system when subjected to a unit step input has an expression for the time response given by**
$c(t) = 0.5 + 1.25\ e^{-3t} - 1.75\ e^{-12t}$
Determine the overall transfer function of the system.

Sol. $\frac{C(s)}{R(s)} = \frac{17.25(s+1.043)}{(s+3)(s+12)}$

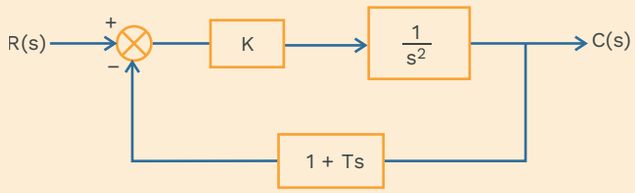
Q20 **The s-domain response of a stable, simple first order system is given by**

$$C(s) = \frac{4A}{s^2} - \frac{A}{s} + \frac{A}{s+4}$$

What is the form of system command in the time domain?

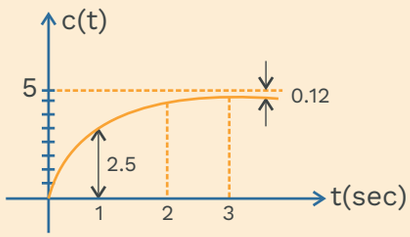
Sol. $r(t) = Bt, \quad t \geq 0$

Q21 **A control system is represented by the block diagram shown below. The maximum overshoot to the unit step input is 20% and time to peak is 2.4 seconds then the ratio of constants K and T will be –**



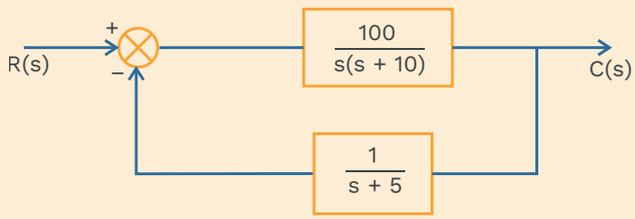
Sol. 3.483

Q22 **The step response of a system is shown below, find the value of forward path gain?**



Sol. K = 40.66

Q23 **For a non unity feedback system, the steady state error for a unit step input is**



Sol. 0